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Experiment: Breadth-First Search (BFS)

Aim

To implement the **Breadth-First Search (BFS)** algorithm using Python to traverse a graph.

Objectives

- To understand the Breadth-First Search (BFS) algorithm.
- To implement BFS using Python.
- To traverse all vertices of a graph level by level.
- To use a queue for graph traversal.
- To analyze the traversal order of BFS.

Algorithm

1. Create a graph using an adjacency list.
2. Select the starting vertex.
3. Mark the starting vertex as visited and add it to the queue.
4. Remove a vertex from the front of the queue.
5. Visit the vertex and display it.
6. Add all its unvisited neighboring vertices to the queue.
7. Repeat Steps 4–6 until the queue becomes empty.

Python Program

```
from collections import deque
```

```
# BFS Function
```

```
def bfs(graph, start):
```

```
    visited = set()
```

```
    queue = deque()
```

```
visited.add(start)
```

```
queue.append(start)
```

```
print("BFS Traversal:")
```

```
while queue:
```

```
    vertex = queue.popleft()
```

```
    print(vertex, end=" ")
```

```
    for neighbor in graph[vertex]:
```

```
        if neighbor not in visited:
```

```
            visited.add(neighbor)
```

```
            queue.append(neighbor)
```

```
# Graph Representation
```

```
graph = {
```

```
    'A': ['B', 'C'],
```

```
    'B': ['D', 'E'],
```

```
    'C': ['F'],
```

```
    'D': [],
```

```
    'E': ['F'],
```

```
    'F': []
```

```
}
```

```
# Driver Code
```

```
bfs(graph, 'A')
```

Sample Output

BFS Traversal:

A B C D E F

Out Put

```
from collections import deque

# BFS Function
def bfs(graph, start):
    visited = set()
    queue = deque()

    visited.add(start)
    queue.append(start)

    print("BFS Traversal:")

    while queue:
        vertex = queue.popleft()
        print(vertex, end=" ")

        for neighbor in graph[vertex]:
            if neighbor not in visited:
                visited.add(neighbor)
                queue.append(neighbor)

# Graph Representation
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
}
```

BFS Traversal:
A B C D E F
=== Code Execution Successful ===

Result

The Breadth-First Search (BFS) algorithm was successfully implemented in Python. The program traversed the graph level by level starting from the given source vertex.

Conclusion

The Breadth-First Search (BFS) algorithm efficiently traverses a graph by visiting all neighboring vertices before moving to the next level. It uses a **queue** to maintain the order of traversal and ensures that each vertex is visited only once. BFS is widely used in Artificial Intelligence, graph traversal, shortest path problems in unweighted graphs, and network analysis.